FT-950 Remote

User & Technical Manual

Yaesu FT-950 web remote



Yaesu FT-950 web remote via the SCU-17 — CAT, WebRTC voice, FT8/FT4, CW skimmer — for a Raspberry Pi.

Version 1.3 · 2026-11-10

github.com/CQ-DJ0SH/ft950-remote

Contents

1 Overview 3
2 Features 4
3 Hardware and radio setup 5
What you need 5
The SCU-17 on the Pi 5
Radio menus 5
4 Installation 6
5 The page 7
6 VFO panel 8
Title bar 8
LCD 8
Tuning 9
TX power and RX · FILTER · LEVEL 9
7 PTT · Memory 11
8 Audio 12
9 DIGI — FT8 / FT4 14
Globe 17
Band conditions (VOACAP) 18
10 RTTY skimmer 19
11 CW skimmer 20
12 JTTY 22
13 HackRF waterfall 23
14 Log 24
15 QSL card by e-mail 25
16 Settings 28
17 Remote power 32
18 Troubleshooting 33
19 Technical reference 34
Backend 34
Files on the Pi 34

1 Overview

FT-950 Remote is a web remote control for the Yaesu FT-950 HF/6 m transceiver. It runs on a Raspberry Pi next to the radio, talks to it directly through a Yaesu SCU-17 interface — CAT at 38400 Bd and USB audio — and serves the whole station as one HTTPS page. No hamlib, no desktop software: every control is mapped to the radio's documented CAT command and checked by reading the value back.

The focus is digital operation (FT8/FT4 with WSJT-X's own decoder) and remote voice over WebRTC, with a CW skimmer, a HackRF panadapter from a second Pi, logging to Wavelog and QRZ.com, and remote power through a Shelly relay.



2 Features

- VFO panel with an LCD modelled on the FT-950's display: FT-950 style analog meter, signal path as lit cells, mode lamps, DSP bars, seven-segment frequency, clarifier, VFO-B, PO/SWR/ALC/ID/AF bars on transmit.
- Title-bar switches: split, ANT 1/2, tuner and ATU tuning, CLAR/CLR, FLT · IPO · R.FLT · AGC, voice profile and FT8 profile.
- Tuning by mouse wheel or drag on a digit, frequency entry, band and mode keys, a frequency shuttle, CTCSS (ENC/TSQ), MON, NAR and a scaled TX power control.
- Push-to-talk in the browser with a watchdog, PTT-LOCK, roger beep, 1750 Hz tone call, MUTE and DROP; 32 memory keys (click recalls, hold 3 s stores).
- Two-way audio over WebRTC/Opus: RX/TX gain, mic AGC, compressor, pre-emphasis, mic echo test, RX recorder with WAV download.
- FT8/FT4 decoding with jt9 on three CPU cores, WSJT-X style sequencer with auto-answer, a queue for several callers to one CQ, waterfall, DXCC and worked-before flags; spots of the stations heard to PSK Reporter.
- HF band conditions under the waterfall from real VOACAP for your locator: 80-40, 30-20, 17-15, 12-10 and 6m in green / yellow / red, with a sun/moon switch for day and night.
- RTTY skimmer: every RTTY signal in the passband decoded at once (45.45 Bd, 170 Hz); transmit as AFSK through the DATA input, with macros.
- CW skimmer: every CW signal in the passband decoded at once; send through the radio's keyer with macros, WPM slider and a browser sidetone (MON); a CW assistant moves a QSO on with one key (QSO or contest style, serial numbers on demand, auto CQ, caller detection, logging).
- JTTY, WSJT-X 3.2+'s keyboard mode: typed like RTTY at any time, with FEC (readable to about -15 dB); every message in the passband, conversation per station, macros, transmit through the DATA input.
- HackRF waterfall relayed from TM-V71 Remote, following the amateur band of VFO A.
- Logging to Wavelog and QRZ.com, auto-log switchable in the LOG title; worked-before index from Wavelog.
- QSL card by e-mail: your picture with the QSO data drawn on it, sent through your own mail account — one QSO, several marked ones or automatically after every log; stations that want no QSL (QRZ.com) are skipped.
- Radio power through a Shelly relay with auto power-off.
- Dark and light theme, 90 % panel zoom, installable as a PWA.

3 Hardware and radio setup

What you need

- Raspberry Pi with Debian 13 / Raspberry Pi OS. A Raspberry Pi 5 is strongly recommended: FT8/FT4 decoding with jt9 is the heaviest job, and the Pi 5 decodes a full band in a fraction of the Pi 4's time. A Pi 4 works, but needs 4–6 s per period at depth D2 instead of 1.5–2 s.
- Yaesu SCU-17 between Pi and radio (USB to the Pi; CAT and RTTY/DATA cables to the radio).
- Optional: a second Pi with TM-V71 Remote and a HackRF One (panadapter), a Shelly relay in the radio's 13.8 V line (remote power).

The SCU-17 on the Pi

The SCU-17 contains a CP2105 dual UART and a TI PCM2903B USB codec. The CP2105's "Enhanced" port (interface 0) carries CAT, the "Standard" port carries PTT/CW on RTS/DTR. The codec is the audio path to and from the radio's DATA jack. Use the stable /dev/serial/by-id path in Settings — it survives re-plugging.

```
/dev/serial/by-id/usb-Silicon_Labs_CP2105_..._if00-port0   CAT, 38400 Bd 8N1
/dev/serial/by-id/usb-Silicon_Labs_CP2105_..._if01-port0   PTT/CW (RTS/DTR)
hw:CARD=CODEC,DEV=0                                        audio in/out
```

Radio menus

- 026 CAT RATE = 38400 Bd (3), 027 CAT TOT = 10 ms (0).
- 031 TX TOT as a safety net for remote operation.
- 050 DATA TX GAIN — transmit level from the DATA jack (FT8, TUNE, voice). Adjust with TUNE until the power is reached and ALC just moves.
- 051 DATA OUTPUT LEVEL — receive level to the Pi (the codec has no capture control).
- 054/055 (DATA shift / VOX) as described in docs/FT8-DECODING.md.

Audio from the Pi reaches the radio on the DATA jack, which the FT-950 takes in the DATA (PKT) modes. Remote voice therefore runs in DATA-U / DATA-L — the VOICE PROFILE sets that up for the current band.

4 Installation

```
sudo apt install wsjtx voacapl    # jt9, ft8code, ft4code; VOACAP
git clone https://github.com/CQ-DJ0SH/ft950-remote /opt/ft950-remote
cd /opt/ft950-remote/backend
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
mkdir -p certs && cd certs && bash ../../deploy/gen-server-cert.sh /path/to/ca
sudo cp ../../deploy/systemd/ft950-remote.service /etc/systemd/system/
sudo systemctl enable --now ft950-remote
```

The service serves https://<pi>:8444/. The certificate is signed by a private root CA (shared with TM-V71 Remote) with a random serial and the Pi's own address; the CA can be downloaded in Settings ▸ General and on the power-off plate. Browsers allow the microphone only over HTTPS, so install the CA once on every device.

Optional nightly reboot that spares a running station: deploy/nightly-reboot.sh (installed as /usr/local/bin/ft950-nightly-reboot, root crontab 00 4 * * *) reboots the Pi only while the radio is off — it skips the night when the FT-950 answers over CAT or its Shelly relay is on, and reboots when the backend does not answer. Each decision goes to the journal (journalctl -t nightly-reboot).

Then open Settings ▸ General: callsign, locator, CAT port and baud rate. Station data, logbook credentials and memory names are stored in backend/runtime.json (mode 0600) and never sent back to the browser.

5 The page



The title bar shows the logo (uploaded under Settings, never part of the repository), your callsign (shown with a slashed zero), the nameplate, the CAT and USB AUDIO lamps, the UTC clock, the theme switch, Settings and — once a Shelly is set up — the power switch.

Below follow the panels VFO, PTT · MEMORY and AUDIO, DIGI, RTTY, CW, HackRF and LOG. Most can be collapsed with ▲; PTT and AUDIO always stay open. The panels are shown at 90 % zoom so the whole station fits on one screen.

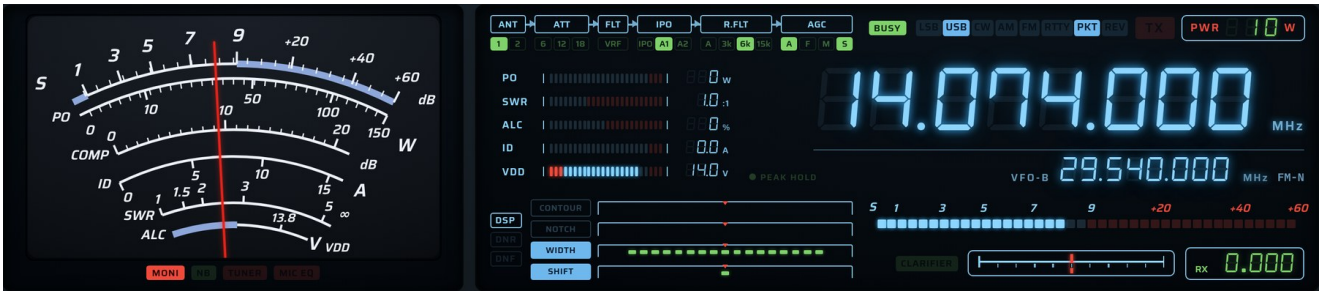
6 VFO panel



Title bar

- LOCK and PLL lamps: dial lock, PLL unlock (a radio fault). A click on LOCK (or Enter/Space) locks or unlocks the dial (LK, verified by read-back).
- SPLIT: receive on VFO A, transmit on VFO B.
- ANT 1 / ANT 2, TUNER (internal ATU in/out), ATU TUNE (starts a tuning cycle — the radio transmits a carrier; confirmation required).
- FLT · IPO · R.FLT · AGC step through VRF/THRU, IPO/AMP1/AMP2, AUTO/15k/6k/3k and OFF/FAST/MID/SLOW/AUTO; the LCD shows the state.
- VOICE PROFILE and FT8 PROFILE: read every value, write only what differs, verify by read-back, report what changed.

LCD



Left window: the analog meter — S on receive, PO on transmit — with the FT-950's scales (S, PO 0–150 W, COMP 0–20 dB, ID, SWR 1–∞, ALC/VDD 13.8 V), white on black with steel-blue bands, and the MONI/NB/TUNER/MIC EQ indicators.

Right window: the signal path ANT · ATT · FLT · IPO · R.FLT · AGC as fixed cells (only the active values light up), BUSY, the mode lamps LSB USB CW AM FM RTTY PKT REV (DATA-U lights PKT + USB, CW-R lights CW + REV), TX lamp and set power, the PO/SWR/ALC/ID/VDD bars, the DSP bars, the frequency to 1 Hz, the CLARIFIER lamp, the clarifier scale (red pointer = offset, ±1 kHz, tall ticks at the ends, a small one every 200 Hz) with its value (the

sign has a fixed place), VFO-B under VFO A and a digital S meter.

The DSP bars CONTOUR · NOTCH · WIDTH · SHIFT are drawn as on the FT-950: a blue bracket with a red pointer in the middle, the lit segments as green dashes. CONTOUR shows two rows with a V-shaped gap at the contour position, NOTCH a dash each side of the notch frequency in two rows; their labels stay dark while the function is off.

The digital S meter under VFO-B is a 40-segment bar, blue to S9 and red above, with the scale 1 … 9, +20, +40, +60 at the needle's S points; it stays dark while transmitting. As on the FT-950's LCD, every 7-segment digit place shows a dimmed 8 and unlit bar segments are dimmed, and nothing changes colour — the PWR box keeps its colours on transmit, the TX lamp shows it.

PO and SWR use approximate calibration points; ID (amps) follows the meter's 5/10/15 A marks. VDD is the supply voltage (RM8, also read while transmitting), approximate: 13.8 V at the meter's 13.8 mark; its bar spans 9.25–15.25 V, 0.25 V per segment. Red zones: the last three segments of PO, ID and VDD, SWR from 2:1, ALC beyond its zone, and the first three of VDD (under 10 V). Peak hold: while transmitting a held segment marks each meter's highest reading; after the transmission the bars and values show these peaks — the values in a green box, PEAK HOLD lit beside it — until the next TX starts (switchable in Settings ▸ Appearance); for VDD it keeps the lowest voltage, the sag under load. CLEAR PH next to NAR drops the held peaks at once (shown only with peak hold on). SWR counts only with some power out. Values keep a fixed number of digits (SWR as 1.5:1), so nothing jumps. The needle moves like a moving-coil meter: damped, rising in about 0.35 s and falling back a little slower. AFU above MHz lights up — green field, dark letters, like BUSY — while VFO A is inside an amateur band of the frequency plan chosen in Settings ▸ Appearance (Germany, IARU Region 1, 2 / USA or 3); its tooltip names the band and its limits.

Tuning

- Mouse wheel or drag on a digit of the LCD frequency.
- Frequency entry with ▲▼ (mic UP/DN), A⇄B and A→B.
- BAND and MODE keys light up at once and confirm when the radio reports back.
- Shuttle next to BAND (◀ DOWN · TUNE · UP ▶): hold left of the centre to tune down, right to tune up; the further out, the bigger the steps (10 Hz, 100 Hz, 1 kHz, 10 kHz) — the label shows the current step.
- CTCSS: off / ENC (send the tone) / TSQ (tone squelch) and the 50 tones, generated by the radio. MON switches the monitor. NAR selects the narrow filters; CLEAR PH beside it clears the held LCD peaks.

TX power and RX · FILTER · LEVEL

The red TX POWER control sets 5–100 W. RX · FILTER · LEVEL opens the groups RX (preamp, AGC, ATT), FILTER (width, shift, squelch SQL 0–255), LEVEL (RF gain in %), CLARIFIER and DSP; FILTER to DSP are equally wide, each value stands right beside its slider. All sliders look as in TM-V71 Remote: filled in the panel colour up to a rectangular fader knob, with a thin ring in that colour (transmit sliders red). The roofing filter is switched with R.FLT in the title bar.

- CLARIFIER: ON switches the RX clarifier, CLR sets the offset to 0; the slider sets the offset (±2.5 kHz in 10 Hz steps, double-click = 0) and switches RX CLAR on. The radio's offset is read back and checked.
- DSP: DNF, NR and NB as switches in one row; below them NOTCH (manual notch, 10–3000 Hz in 10 Hz steps) and CONTOUR (position 1–30), each with a switch, a slider and its value. The slider only sets the frequency or position, the switch turns the function on and off. Contour gain and width stay in the radio's menu (069/070).

7 PTT · Memory



- PTT: hold the button or the space bar. The browser renews a lease every second — if the page goes silent, the Pi drops PTT after 2.5 s. Maximum 180 s per over. The microphone is muted unless you transmit.
- PTT-LOCK latches transmit; ROGER adds a roger beep; 1.750 kHz arms the repeater tone call: the next PTT sends 1750 Hz instead of the microphone, releasing PTT disarms it (FM modes).
- MUTE mutes the receive audio (descending chime), DROP blanks only the current over and lifts when the station stops (ascending chime).
- PTT → VFO shows which VFO transmits. In the installed PWA, level bars for RX and MIC stand left and right of the PTT button.
- Memory keys M1–M32 (2 × 8 per side): click recalls frequency and mode to VFO A, in FM with CTCSS, plus VFO B and split if the key has them; hold 3 s to store VFO A in that channel (confirmation before overwriting).

Settings ▸ Memories edits all 32 channels: frequency, mode and CTCSS (mode and tone) go into the radio with MW, verified by MR; split, VFO B and a name are kept on the Pi, because the FT-950's memories have none of them. DEFAULTS (M1-M32) fills one band per key row for 80/40/20/15/10/6 m — left SSB | CW calling frequencies, right FT8 | FT4 — then 10/6 m FM, the 10 m repeater (split −100 kHz), SSTV and the IARU Region 1 emergency centres.

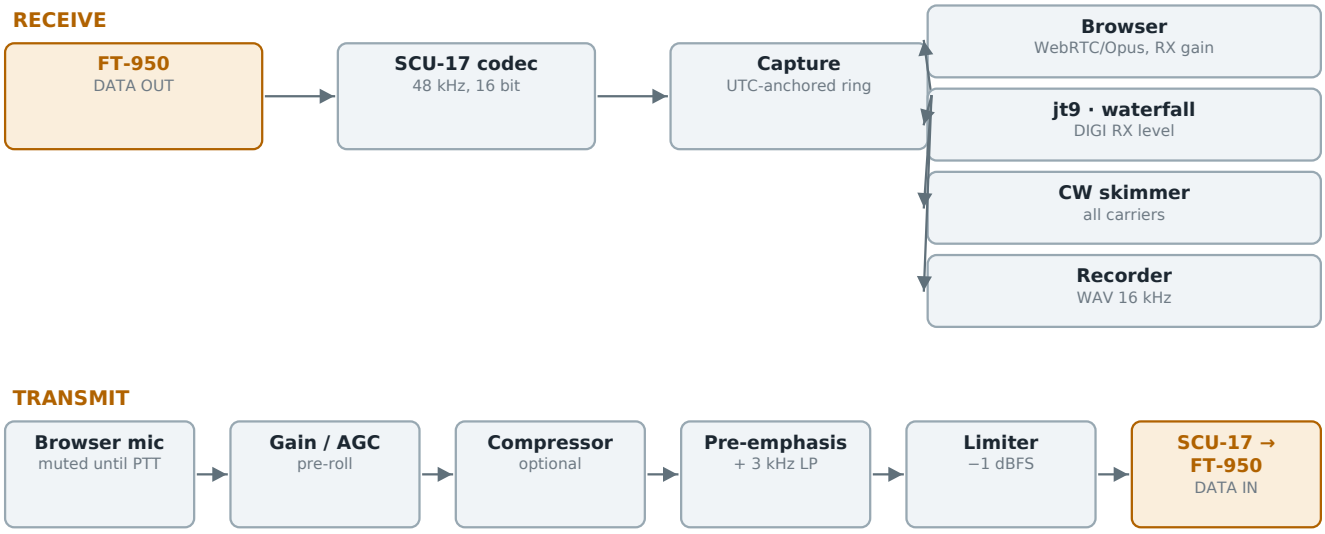
8 Audio



CONNECT starts WebRTC: the radio's audio plays in the browser, the microphone goes to the radio while you transmit. RX GAIN and TX GAIN set the levels. The switches in the title row: AGC rides the microphone toward a target level; MIC TEST records your microphone through the whole path and plays it back without transmitting — at listening level, the meters keep showing the real transmit level. On a desktop the browser's own microphone processing stays on (switching it off drops the level by about 40 dB on macOS); a built-in microphone is preferred to a Bluetooth headset.

REC records the receive audio on the Pi (up to 30 minutes, 16 kHz), ▶ plays it, ↓ WAV downloads it. Settings ▸ Audio adds a compressor and pre-emphasis for SSB voice, the FT8 TX drive, the codec output level and the FT-950's audio menus with saved defaults.

The graph is split: on top the levels of the last 30 s, below the audio spectrum 0–3 kHz (the FT-950's SSB/DATA passband) — receive audio in violet with a yellow peak hold that falls away in about 10 s, your microphone in red while you transmit or run MIC TEST. PEAK in the legend switches the peak hold. The spectrum is computed in the browser and starts with CONNECT.



FT8/FT4 and TUNE: tones generated on the Pi (ft8code/ft4code) go straight to the codec; CW is keyed by the radio itself (keyer memory).

9 DIGI — FT8 / FT4



DECODE in the panel's title bar switches decoding on and off (as in the CW skimmer); off, the waterfall stops and is cleared (the Pi then computes no waterfall either). AUTO picks FT8 or FT4 by the nearest standard frequency — on 80 m FT8 (3.573) and FT4 (3.575 MHz) are only 2 kHz apart. Decoding runs every period with WSJT-X's jt9 — or a faster build chosen in Settings ▸ FT8 ▸ Decoder (see below); the WSJT-X plate in the title bar shows the one that runs (2.7, 2.7 fast, 3.3 MT). The passband is split over three cores. D1/D2/D3/AUTO in the footer pick the depth: D2 (default) puts the decodes on screen about 1.5–2 s after the period on a Pi 5, D3 finds a few more weak signals in about 3 s, AUTO follows the measured run time. Received messages list SNR, time offset, frequency, message and DXCC, with worked-before colouring.

Title bar: DECODE, CQ with its target, Auto Answer, Auto Hunt, the WSJT-X plate, the next transmission, Halt TRX and the TX lamp. Strip under the tables: mode (AUTO · FT8, FT8, FT4), the cycle bar (RX or TX and the seconds into the period), EVEN/ODD, DECODE with the run time and the number of decodes of the last pass, the NTP LED and the PSK Reporter LED, then RX with the receive level and TX with TUNE. Footer: dial frequency and power (framed), decode depth, OVL (codec overload), the TX frequency with Adopt Frequency, palette, contrast and brightness of the waterfall, and the VOACAP band conditions.

- Click a decode to put the RX window (green frame, if switched on in Settings ▸ FT8) there. With Adopt Frequency on (footer, next to the TX frequency) the TX box (red) moves too, and Auto Answer and Auto Hunt answer on the partner's frequency — the TX box follows while their QSO runs. A click in the waterfall sets TX by hand (the click is the middle of the box, so it sits over the signal clicked; the TX frequency itself stays its lowest tone, as in WSJT-X) and switches Adopt Frequency off, so the next station does

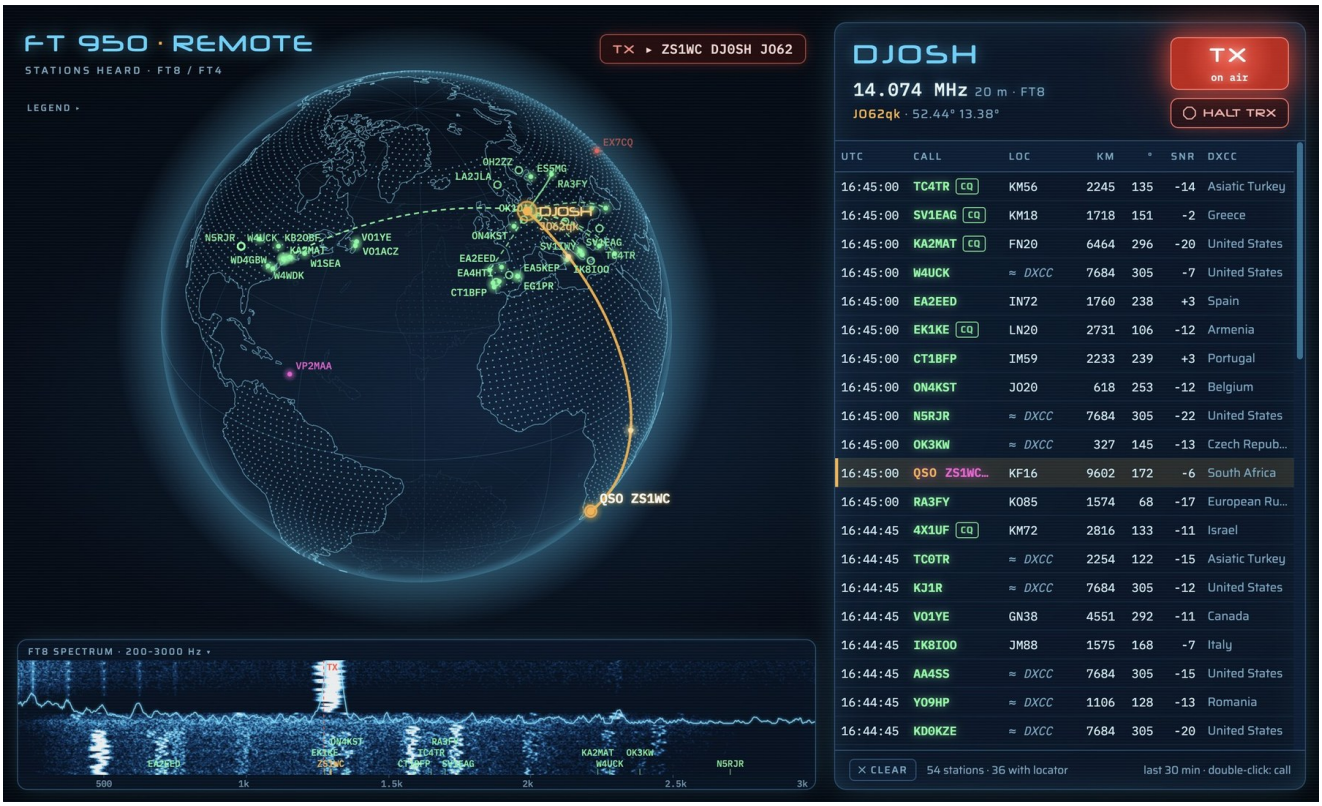
not pull it away; Shift+click moves only the RX window. double-click answers the station — in Received and in My Messages — and selects its session in Sessions, so Session Messages shows that QSO. A double-click within the first ~1.7 s of your own TX period still transmits in that period (start at once, the partner sees a slightly positive DT); later it goes out in the next one.

- Palette Hologram (dark blue to cyan, as the globe) adds a callsign strip at the foot of the waterfall: every decoded station of the last period with a tick at its frequency, in up to four rows. The waterfall gets shorter by that strip, so the labels never cover the signals. Hovering a callsign shows country, SNR and the message; a double-click calls the station, as in Received. Settings ▸ FT8 ▸ Callsigns (all palettes) shows the strip with every palette.
- The filters above Received Messages show all, CQ only, RR73 (with RRR) only or 73 only. The arrow icon ↓ in each list head (follow) keeps the newest row in view, Session Messages included.
- A failed session shows just error in the list; the reason comes as a toast (when it happens and on a click on the row) and as the tooltip.
- With a session marked (click its row), QRZ Visit opens the station's QRZ.com page and Map Display a world map: position from the locator (without one, the centre of the DXCC entity), great circle, distance and bearing from your locator. Both open a popup window.
- CQ calls CQ — to everyone (–) or directed with the choice next to it: DX, EU, NA, SA, AF, AS, OC, POTA or SOTA (CQ DX DJ0SH JO62). Auto Answer: while idle, a station calling you is answered automatically.
- My Messages: on your TX lines the last column (DXCC/CNT) counts the attempts — 2/5 for a QSO message (five unanswered repeats end the sequence), 3/40 for CQ; the last attempt is amber.
- DX Call in the My Messages head calls a station typed in: call, grid (optional; 6 characters are cut to 4, an invalid one is dropped), period auto / even / odd, then Call DX. auto answers the station's last decode if it was heard, otherwise it calls in the next period. Your own call is refused; a click on a decode fills the fields.
- Auto Hunt calls new stations by itself: while no QSO runs, after every full decode pass it calls the best new CQ (a plain CQ, not CQ DX/NA/…; new DXCC first, then the strongest) and waits for the QSO to end. A station that never answers is struck through in Received Messages and not called again (the list survives restarts). While on, the button is red and shows Hunt and the time left as m:ss (∞: no switch-off). It calls only while a browser shows the DIGI panel. Whom it calls, the minimum SNR, the switch-off time (default 180 min, 0 = never) and the no-reply list are in Settings ▸ FT8.
- Hunt and the two periods: after a QSO, Hunt calls from the first period it hears — the one opposite to our TX — and so keeps to one parity for long runs. Settings ▸ FT8 ▸ “Pick only after hearing both periods” (default off) makes it listen one period more (about 15 s per QSO) and pick the best station of both, even and odd.
- Timing: the next message is decided about 0.7 s into the period and goes on the air at about +1 s (the latest start is +2 s). An early pass at 13.4 s searches the whole band, split over three cores (~0.6 s), while the other station is still sending — callers, a partner who moved and an empty period are known before our next start. Right after the period a narrow partner pass (~0.4 s) checks the partner's frequency; the full pass (~1.5 s) fills the tables. Measurements: docs/FT8-DECODING.md §13.

- SKIP in the Sessions header (amber while a QSO runs) ends the QSO in progress at once: TX to that station stops, and the rest goes on — CQ calls again (or takes a waiting caller), Auto Hunt and Auto Answer go to the next station. A station Auto Hunt was calling goes on the no-reply list, so it is not picked again straight away; the session shows as skipped. HALT, by contrast, stops everything.
- The sequencer follows WSJT-X: call, report, R-report, RR73, 73; it can log automatically: Auto-log in the Sessions header (default on) logs every completed QSO to Wavelog, and Settings ▸ Log can send it to the QRZ.com logbook as well (its own switch, default off); with neither it is offered in the log form. A session counts as logged when every target accepted it. Answers that differ from the textbook: RR73 or RRR while we still repeat our report is answered with 73 (their RR73 says our report arrived), “R <locator>” (WSJT-X's contest-style acknowledgement) counts like an R-report and is answered with RR73 (RST received stays empty), and a bare 73 completes the QSO only once the reports went across — sent too early, it is ignored, so nothing is logged or carded half-done. A new QSO the sequencer starts by itself (Auto Hunt, Auto Answer, a caller to our CQ) is selected in Sessions, so Session Messages follows it; a session clicked afterwards stays until the next one starts. While a QSO runs, Session Messages also lists the partner's messages to others (their CQ, their QSO with someone else) in grey — they change nothing in the sequence; so is the decode that started the session when it was not addressed to you. Our own lines appear as soon as the next message is decided, about a second before the radio keys: TX ▷ dimmed while pending, TX ▶ once on the air, TX ✕ struck through if it did not go out (too late for the period, or halted). A reply that comes in while our older message is still on the air is answered next period — the QSO is not closed before our own 73. While a QSO runs, a narrow decode around the partner (FT8 ±200 Hz, FT4 ±300 Hz — FT4 needs the wider margin) has their reply within ~0.1 s of the period end, so the next message usually starts 0.5–1 s into our period; without a reply there it repeats at once. It can send RRR instead of RR73 (Settings ▸ FT8).
- Several callers to one CQ: the first goes to a QSO, the others wait in a queue and are answered one after the other when the QSO ends, instead of the next CQ. Order: new DXCC, DXCC new on this band, not worked on this band, then the strongest. A waiting station drops out after 2 minutes without a new call, or as soon as it is heard with someone else or calling CQ; one just worked does not queue again. Waiting stations are listed in italics at the end of Sessions (next, waiting 2, …; the tooltip shows their message and SNR); a click puts one at the front.
- A caller decoded while our CQ is already on the air: the CQ is cut and the answer goes out in the same period, as long as it can still start in time (FT8 up to 1.7 s into the period, FT4 1.2 s); later it goes out in the next period. A cut CQ does not count towards the CQ limit.
- Calls FT8 cannot carry in full (DL1ABC/QRP, W1AW/4, PA/DJ0SH — only /P and /R fit) go out as WSJT-X sends them, in short (hashed) form: <DL1ABC/QRP> DJ0SH JO62. The other station recognises its own call; your call, grid and reports stay in full.
- PSK Reporter (Settings ▸ FT8): every station decoded here is reported to pskreporter.info, with your call and locator as the receiver. The PSK Reporter LED in the strip is green after a good upload, red when the last one failed, dark until the first; its tooltip has the details. While the reporting is off the LED is hidden, its place kept. Below 1450 px window width the label shortens to PSK.

- RX in the strip sets the DIGI RX level on the Pi (software gain −20 … +20 dB, double-click: 0 dB); the meter right of it shows the receive level, target −40 … −20 dBFS. TX sets the FT8/FT4 TX drive (as FT8 TX DRIVE in Settings ▸ Audio); TUNE next to it sends a steady tone on the TX frequency and counts down the seconds left. Halt TRX, right in the title bar, is active during an FT8/FT4 transmission, a sequence or while Auto Hunt is on (dimmed otherwise); it stops at once and switches Auto Hunt off. NTP in the strip is an LED: green with the clock synchronised, red without (no decoding).

Globe



Globe, beside the title of Received Messages, opens a popup with the stations heard on a slowly turning, see-through hologram globe (land as an even dot grid that looks the same from any side, the poles included); your own station stands out in amber with call and locator. Stations of the last 30 minutes are points with their call (where they would overlap, the important ones win): magenta a new DXCC, green new on this band, red worked; a hollow ring sits at the centre of the DXCC entity when the station sent no locator. A call that appears for the first time draws a line out from you that fades after about 20 s; stations calling CQ now (last 45 s) have a dashed line, the QSO in progress an amber line with pulses running along it.

- The table beside it has your call as its title, the frequency with band and mode, a TX lamp (red and glowing while transmitting, with the message that goes out) and HALT TRX, as on the main page. Columns: UTC, call (with a CQ tag while it calls), locator, km, bearing, SNR, DXCC.
- A click on a row (or a point) turns the globe to the station and stops it there; RESUME, bottom left, turns it on again - also after turning it by hand with the mouse. The wheel zooms. A station clicked on the globe is marked in the table too (cyan stripe) and scrolled into view; the table keeps it there while it re-sorts, until you scroll it yourself.
- While transmitting, a small TX line at the top right of the globe shows what goes out

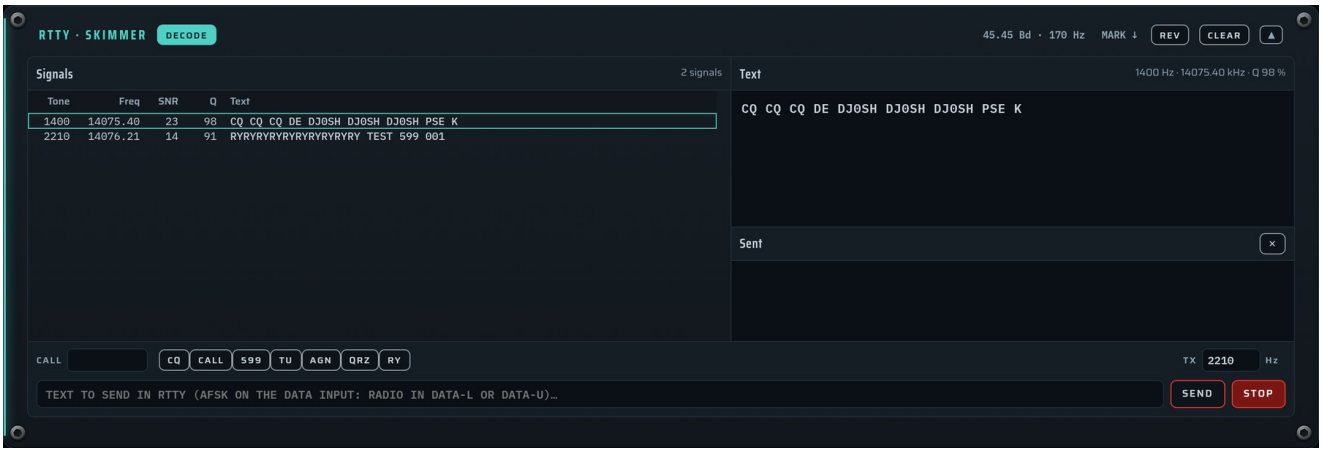
(TX ▸ message, TUNE, VOICE or PTT). During a QSO it shows the partner's last message above it (RX ◂ message and SNR) and stays through the receive periods too, your last message then dimmed.
- ✕ CLEAR in the table footer empties table and globe; only stations heard afterwards come back. GRAY LINE beside it (amber when on, remembered) shades the night side from the sun's position now — twilight to −12°, the day side faintly lit — with the day/night border as a dashed amber line: DX along it often goes well.
- A double-click on a row calls the station - through the main page, exactly as a double-click in Received Messages (RX window, Adopt Frequency, log form, then the call). Without the main page the popup does not call.
- Under the globe runs the FT8 spectrum across the full width: a waterfall over the decoder's passband (200-3000 Hz from Settings ▸ FT8, the DIGI waterfall's data) in the hologram colours with the current spectrum over it, the stations of the last period with their call at their audio frequency (only while the legend is open — folded, the strip stays clean), the QSO partner in amber and your TX frequency as a red line while a sequence runs. With DECODE off it is empty. Legend and FT8 spectrum fold with their header line (▾/▸); the browser remembers it.
- The popup follows the main page's light or dark theme (☀/☾), also while it is open.

Band conditions (VOACAP)

At the bottom right of the panel a sun or a moon and five chips — 80-40, 30-20, 17-15, 12-10, 6m — show how the bands are open from your locator, in green, yellow or red, as iDigi shows them. The values are computed on the Pi with VOACAP (voacapl): VOACAP predicts a path, not a place, so the question goes to a ring of 36 paths around you (12 directions × 1500 / 4000 / 9000 km) at 100 W FT8 with an isotropic antenna. Per band the reliability is the mean over the ring, and a pair is as good as its better band: green from 0.6, yellow from 0.3, red below. Hovering a chip shows both bands in per cent and the trend of the next three hours.

- Sun or moon follows the real time of day at your locator. A click shows the other half of the day — the mean over today's daylight or darkness hours there — with a framed switch and dashed chips; a second click goes back to now.
- Updates: one run covers all 24 UTC hours of the month's median ionosphere (about 0.5 s, at low priority so jt9 never waits). It is computed once per UTC day and again when the locator or the sunspot number changes. The sunspot number is NOAA SWPC's latest monthly value, fetched once a day; the last good one is kept for an offline Pi. The page asks every 5 minutes, so the next hour, sunrise and sunset show within 5 minutes.
- Limits: VOACAP is a monthly median of the F2 layer. It does not see flares or geomagnetic storms, and on 6 m it shows only the rare F2 openings of a solar maximum — not sporadic-E, which opens 6 m in a European summer. A red 6m chip does not rule out Es.
- Without voacapl, or without a locator in Settings ▸ General, the strip is not shown.

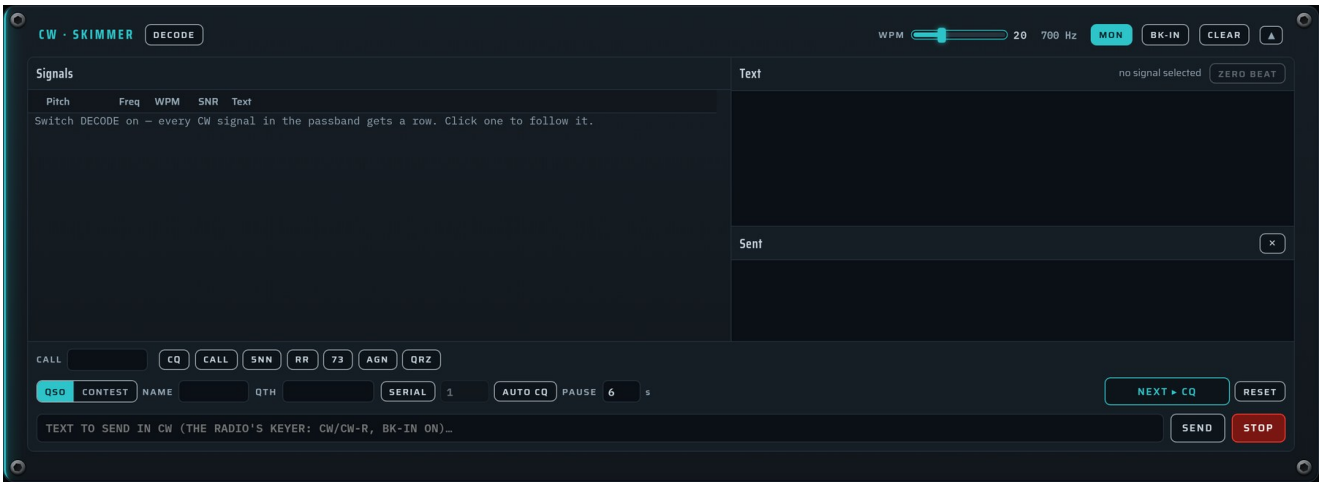
10 RTTY skimmer



The RTTY panel below DIGI decodes every RTTY signal in the passband at once — standard amateur RTTY: 45.45 baud, 170 Hz shift, Baudot with US figures, 1.5 stop bits. DECODE switches it on (it runs on the Pi only while on). Each signal gets a row: tone (the audio centre between mark and space), frequency on the air, SNR, quality (share of characters with a correct stop bit) and the running text; a click shows its whole text on the right and takes its callsign into CALL.

- How it works: every 0.5 s a spectrum of the last 2 s is searched for pairs of peaks 170 Hz apart with a dip between them (up to 8 signals, slow drift is followed). Per signal, mark and space go through a filter matched to the bit; their levels are compared after each is normalised to its own peak, so fading of one tone does not flip the bits. A UART finds start bit, five data bits and stop bit. A character only counts where one tone stands clearly above the other, so noise prints nothing, and a row is shown only once most of its frames are clean.
- Mark and space: on the air the mark is the higher frequency. In LSB / DATA-L that is the lower audio tone, in USB / DATA-U the higher one — the panel follows the radio's mode (MARK ↓ / ↑). REV swaps them, for a station sending reversed (receive and transmit).
- Transmit: AFSK through the DATA input, like FT8 — the radio in DATA-L (usual), DATA-U or USB, at the FT8 TX drive. The text goes into the send line, by hand or from the macros CQ, CALL, 599, TU, AGN, QRZ and RY ({MY}, {LOC} and the CALL field are filled in); Enter or SEND transmits it with CR LF before and after, mark idle in front. TX is the centre frequency in Hz of audio (300–2600); a click on a signal sets it to that station; DEFAULT puts it back to 2210 Hz (mark 2125 / space 2295 Hz). STOP and Halt TRX stop at once; a watchdog ends the transmission at the latest 3 s after its length. While the FT8 sequencer or Auto Hunt runs, RTTY does not transmit.
- Own transmissions in Text: each appears as a red TX block in the received text of the station at the TX frequency, at the point it went out — the QSO reads in order and the block stays in place while the text scrolls. Sent keeps the list. While we transmit (and 0.6 s after) the RTTY, CW and JTTY receivers hear silence: the radio puts our own signal on its audio out, and it would come back as an echo of the TX text.
- The panel can be hidden and gets its own accent colour in Settings ▸ Appearance.

11 CW skimmer



DECODE starts the skimmer. Every CW carrier in the passband gets its own decoder; the table shows pitch, RF frequency, speed, SNR and the running text. Click a row to follow it: its text appears under Text, ZERO BEAT tunes VFO A onto it.

Sending uses the radio's own keyer: the text is written to keyer memory 5 and played (KM/KY), in chunks of 50 characters. The radio must be in CW or CW-R with BK-IN on. Macros (CQ, CALL, 5NN, RR, 73, AGN, QRZ) fill the send line; CALL is taken from the selected signal. Own transmissions show as red TX blocks in the received text of the signal on the radio's pitch (the one we are zero-beat with), as in RTTY; while we send, the skimmer hears silence (no echo).

- WPM is a slider (4–60) in the panel head; it sets the keyer speed of the radio (KS) as soon as you let go, and follows the radio every 10 s. Each time the service connects to the radio it starts at 20 WPM (FT950_CW_START_WPM in .env, 0 keeps the radio's speed). The pitch beside it is read from the radio (KP).
- Before each send the panel switches the radio's keyer on (KR1) and sets CW memory 5 to TEXT (menu 023), both checked by read-back. The text is played with KYA - on the FT-950, text written with KM plays with KY6…KYA (memory 1…5), not KY1…5. A message recorded with the paddle in memory 5 is therefore not available.
- MON plays your own CW in the browser: the radio's sidetone and monitor reach only its speaker, not the Pi. The page keys a tone at the radio's speed and pitch, paced like the radio's chunks, with soft edges (no clicks). It is made in the browser and never transmitted; STOP ends it with the radio, MON off at once.
- If something does not go out, GET /api/cw/diag shows (read only) the keyer, memory 5 type, both key jacks and the content of memory 5.

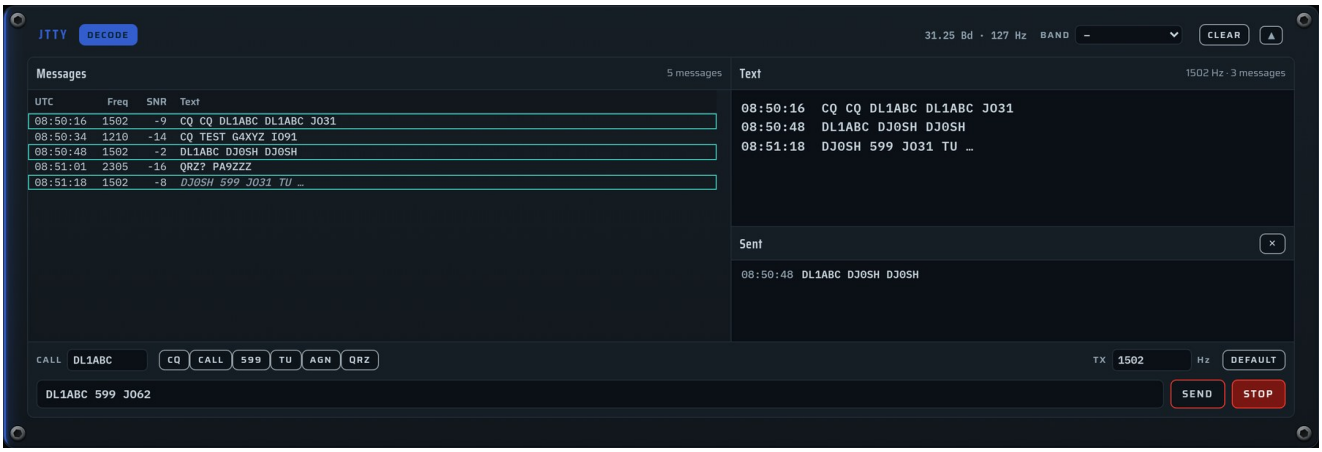
The CW assistant (the row above the send line) moves a QSO on with one key: NEXT, or Enter in an empty send line or in CALL. The button names the step it sends.

- Calling CQ: CQ → REPORT → TU + LOG, then CQ again. After your CQ, a call heard near your own pitch (where a caller zero-beats) comes into CALL, framed — it is only answered when you press NEXT. What the other station sends is read from its text and shown as RCVD: RST, serial number (cut numbers such as 1T2 = 102), name and QTH; it goes into the log.
- Answering a station: a double-click on a signal tunes onto it (zero beat), takes its call

into CALL and sends ANSWER; then REPORT and 73 + LOG (contest: REPORT + LOG).

- QSO or CONTEST: QSO sends the full exchange with RST 599, NAME and QTH (stored in the browser); CONTEST the short one (CQ TEST, CALL 5NN, TU).
- SERIAL adds a serial number to the report on demand; the field shows the next one (editable), it counts up after every logged QSO.
- AUTO CQ repeats CQ with a listening PAUSE until a station answers, STOP or RESET — at most 30 calls. It runs in this browser: closing the page stops it.
- The log entry goes to Wavelog and QRZ.com as mode CW with RST sent and received, name, QTH, start and end time, with SERIAL also STX/SRX.

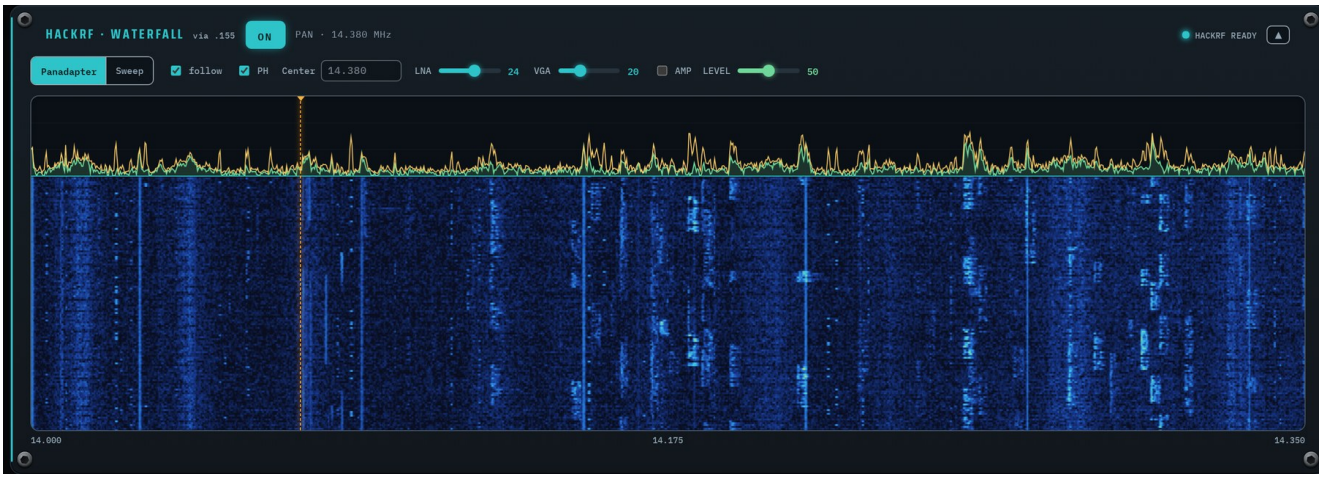
12 JTTY



JTTY is the keyboard mode of WSJT-X 3.2 and later, by Joe Taylor K1JT, Steve Franke K9AN, Rob Sherwood G4KLA and the WSJT-X team (GPL v3). It is worked like RTTY — both sides type and send whenever they like, there are no T/R periods — but with forward error correction: 4-GFSK at 31.25 baud, 127 Hz wide, frames of 1.89 s, readable down to about −15 dB where 45.45-baud RTTY is long gone. Calls, reports and locators travel in compact form.

- DECODE starts jttyd on the Pi (decoder/jttyd, built against WSJT-X 3.3's decoder). It first sets up its tables on a test signal of its own (STARTING, about 25 s), then listens over the whole passband — about 12 % of one core.
- Every message is a row: UTC, audio frequency, SNR, text; one still arriving is shown in italics with … . A click follows that station: Text shows its conversation (every message on its frequency), CALL takes its call, TX its frequency. Our own transmissions join the list and the conversation as TX lines in red (“sending …” while on the air); received ones are labelled RX. CALL is taken only from what the other station sent.
- Sending: the send line or the macros CQ, CALL, 599, TU, AGN, QRZ (kept short, so JTTY's compact atoms carry them); Enter or SEND transmits at once — no waiting for a period. The radio must be in DATA-U or USB; the drive is the FT8 TX drive. STOP and Halt TRX end it; while the FT8 sequencer runs, JTTY does not transmit. DEFAULT puts TX on 1500 Hz, the middle of the calling range.
- BAND tunes VFO A to a JTTY dial frequency in DATA-U (WSJT-X 3.3's list, preliminary): 1.839, 3.572, 7.078, 10.140, 14.090, 18.104, 21.090, 24.920, 28.090, 50.160 MHz. 20 m at 14.090 MHz is the suggested first meeting place.
- WSJT-X's decoder listens on the receive frequency and 1200–1800 Hz; jttyd sets its tolerance to the whole passband, so every JTTY signal is found — measured without loss of sensitivity.

13 HackRF waterfall



The HackRF One sits on the second Pi with TM-V71 Remote. FT-950 Remote relays its waterfall (Settings ▸ General ▸ HackRF source). With follow on, the view shows the whole amateur band of VFO A; one click tunes VFO A there. The HackRF is shared: what you set here also applies to the TM-V71's waterfall.

In the band view TM-V71 Remote sends a fine row of just that band (8192-point FFT, 244 Hz per point, 4 blocks averaged) — 820 points for 40 m instead of about 50 from the normal 2 MHz row. The mouse shows frequency and level under the pointer.

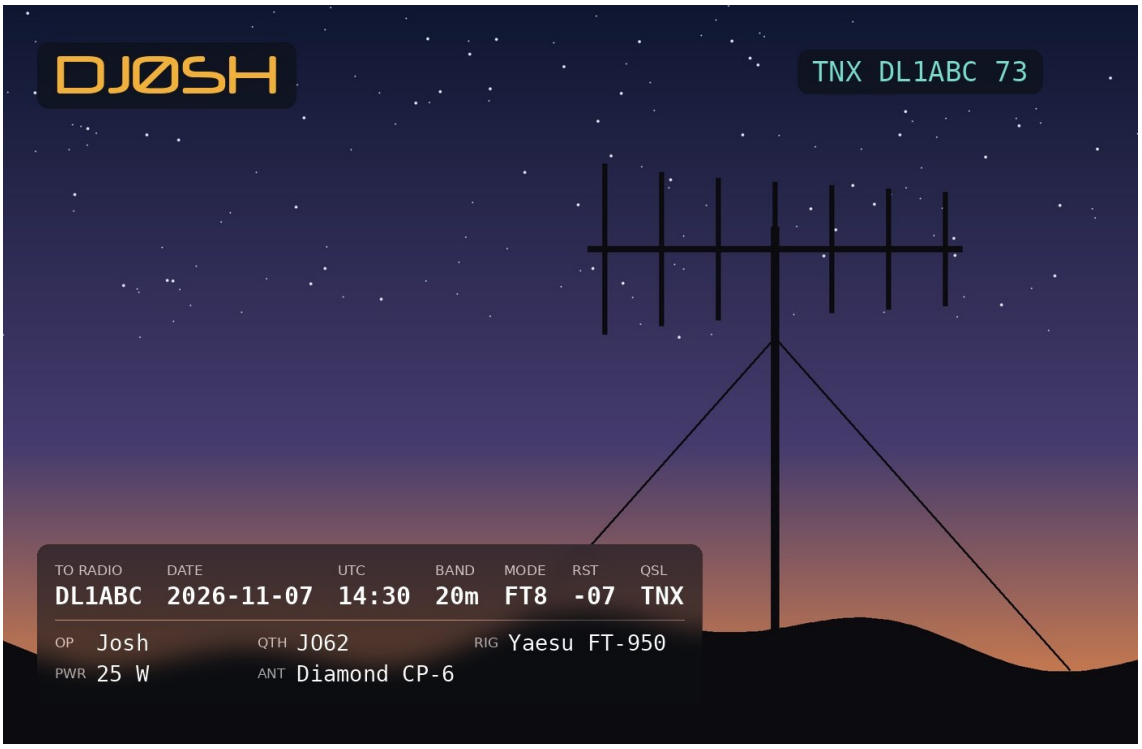
14 Log

The LOG panel logs to Wavelog (API key and station profile) and the QRZ.com logbook — each with its own button beside the callsign: LOG WAVELOG and LOG QRZ.COM (the form stays, so a QSO can go to both; a ✓ marks where it went). QRZ LOOKUP fills name, QTH and locator; NEW STATION / WORKED shows in the title bar. The worked-before index is downloaded from Wavelog and colours the FT8 list. Settings ▸ Log holds the accounts — the keys never leave the Pi.

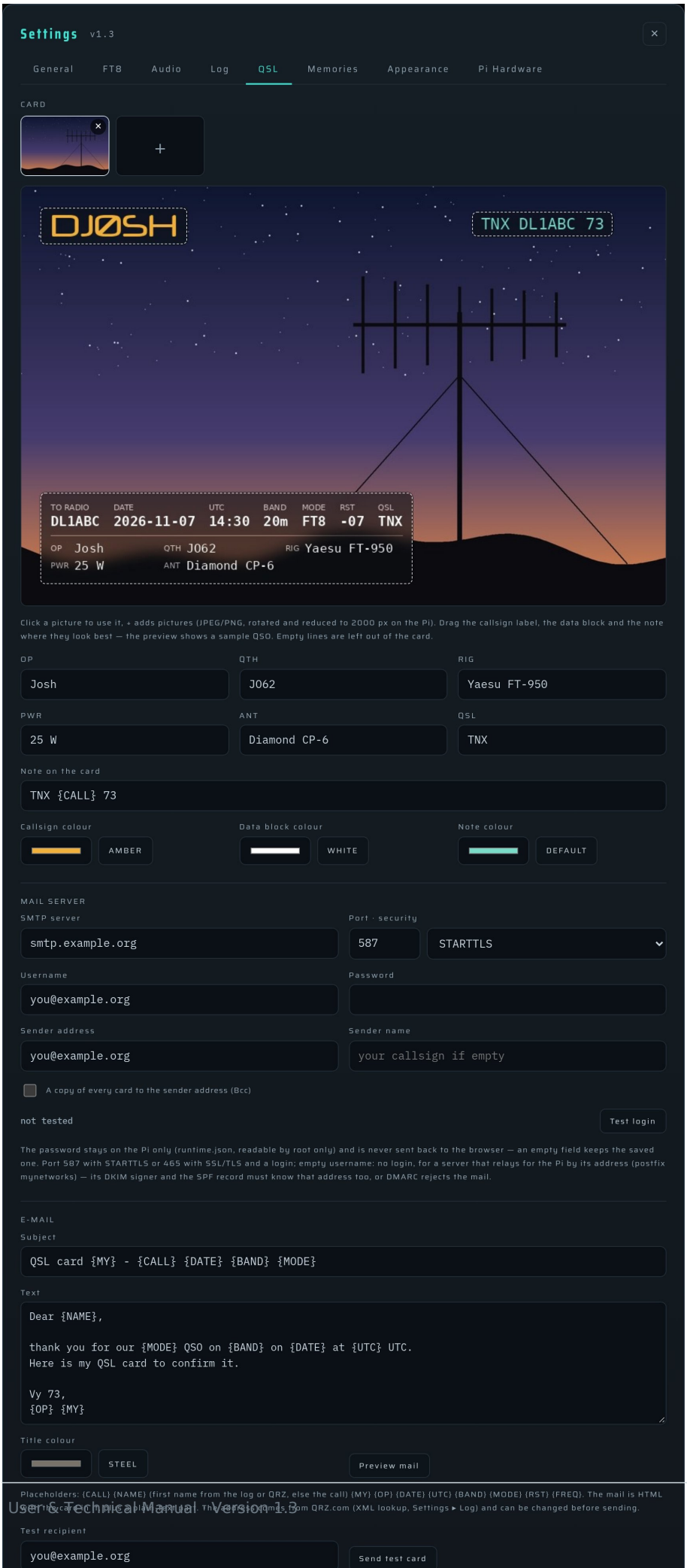
In the title bar, LEDs as in the page's own title bar: WAVELOG and QRZ green when set up. Three switches (filled = on): AUTO WAVELOG logs QSOs completed by the sequencer to Wavelog (as Auto-log in Sessions and Settings ▸ Log), AUTO QRZ to the QRZ.com logbook as well, AUTO QSL sends the QSL card by e-mail after every successful log (chapter 15). The LOG buttons are only outlined while the form is empty and filled once a callsign is in it — then there is something to log.

Recent QSOs: a click marks a row, Ctrl/Cmd-click adds or removes one, Shift-click marks a range. QRZ Visit, Map Display and QSL Card (shows the card) work on the last clicked row, Send QSL on all marked ones. The list follows changes made on the Pi by itself (checked every 5 s): QSOs logged automatically and QSL cards sent by AUTO QSL appear at once, the latter with a notice. The QSL column shows ✓ sent, ✗ not sent, ⊘ skipped — hover for why.

15 QSL card by e-mail

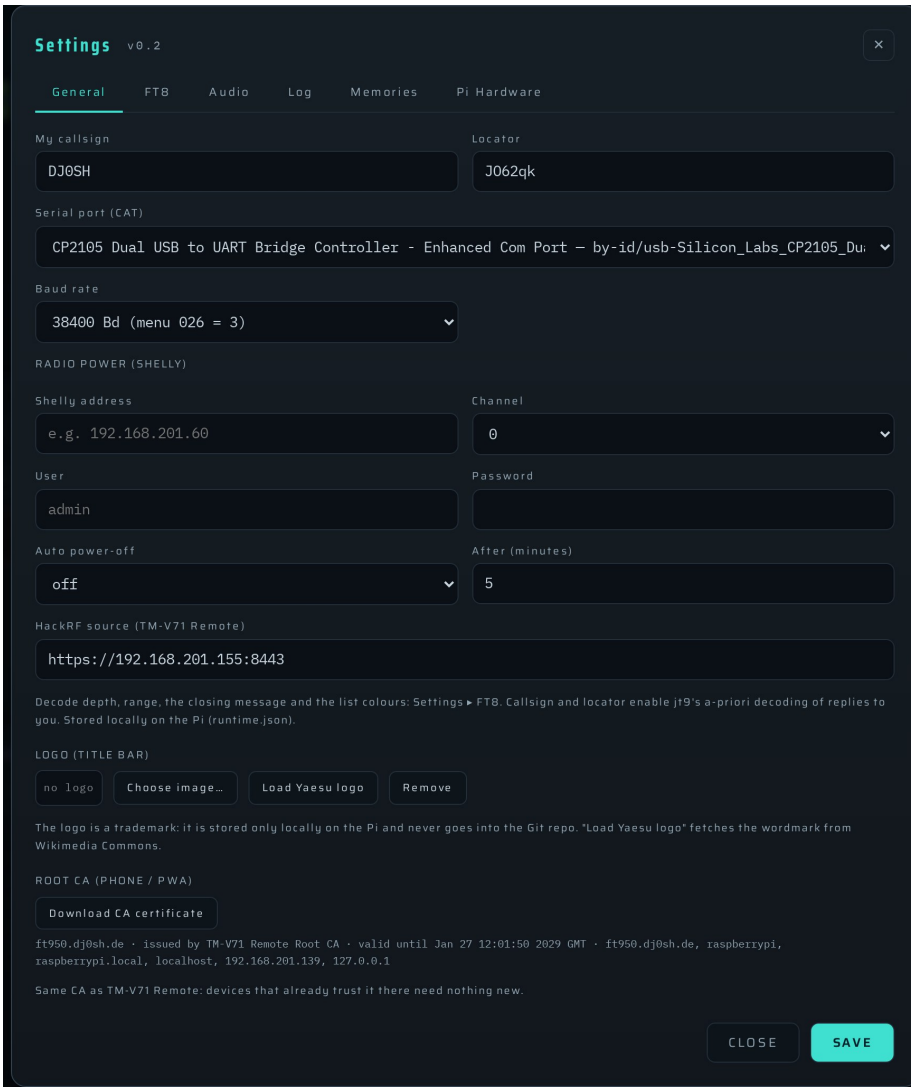


A QSL card is your own picture with the QSO data drawn on it on the Pi: your callsign as a label (in the nameplate font, 0 as Ø), the data block (to radio, date, UTC, band, mode, RST, QSL and below it OP, QTH, RIG, PWR, ANT side by side) and a note of its own. The mail is HTML with the card in it, the page's title plate in the header (steel title bar, both calls in amber), a link to this project and a plain-text part; it goes out through your own mail account. In the HTML part your own call is written with Ø as on the card; the plain-text part keeps the 0, so the call can be copied.


Settings v1.3
General FT8 Audio Log QSL Memories Appearance Pi Hardware
CARD
DJ0SH
TNX DL1ABC 73
TO RADIO DATE UTC BAND MODE RST QSL
DL1ABC 2026-11-07 14:30 20m FT8 -07 TNX
OP Josh QTH JO62 RIG Yaesu FT-950
PWR 25 W ANT Diamond CP-6
Click a picture to use it, + adds pictures (JPEG/PNG, rotated and reduced to 2000 px on the Pi). Drag the callsign label, the data block and the note where they look best — the preview shows a sample QSO. Empty lines are left out of the card.
OP
Josh
QTH
JO62
RIG
Yaesu FT-950
PWR
25 W
ANT
Diamond CP-6
QSL
TNX
Note on the card
TNX {CALL} 73
Callsign colour
AMBER
Data block colour
WHITE
Note colour
DEFAULT
MAIL SERVER
SMTP server
smtp.example.org
Port · security
587
STARTTLS
Username
you@example.org
Password
Sender address
you@example.org
Sender name
your callsign if empty
A copy of every card to the sender address (Bcc)
not tested
Test login
The password stays on the Pi only (runtime.json, readable by root only) and is never sent back to the browser — an empty field keeps the saved one. Port 587 with STARTTLS or 465 with SSL/TLS and a login; empty username: no login, for a server that relays for the Pi by its address (postfix mynetworks) — its DKIM signer and the SPF record must know that address too, or DMARC rejects the mail.
E-MAIL
Subject
QSL card {MY} - {CALL} {DATE} {BAND} {MODE}
Text
Dear {NAME},
thank you for our {MODE} QSO on {BAND} on {DATE} at {UTC} UTC.
Here is my QSL card to confirm it.
Vy 73,
{OP} {MY}
Title colour
STEEL
Preview mail
Placeholders: {CALL} {NAME} (first name from the log or QRZ, else the call) {MY} {OP} {DATE} {UTC} {BAND} {MODE} {RST} {FREQ}. The mail is HTML with a text fallback. {NAME} comes from QRZ.com (XML lookup, Settings ▸ Log) and can be changed before sending.
Test recipient
you@example.org
Send test card

- Settings ▸ QSL ▸ Card: + adds pictures (rotated and reduced to 2000 px on the Pi), a click picks one. The preview is drawn by the Pi exactly as the card will be; drag the callsign, the data block and the note into place. The note takes placeholders, e.g. TNX {CALL} 73; empty lines are left out.
- Colours: Callsign colour (the label on the card and both calls in the mail header, default amber), Data block colour (values; the labels and the divider take it fainter, default white) and Note colour (default turquoise) — each with a reset button; the preview follows at once. The other station's call is never wider than 10 characters (DJ0ABC/QRP): a longer one is drawn smaller, so the data block does not grow into the other blocks.
- Mail server: SMTP server, port and security, username and password (stays on the Pi, never sent back to the browser), sender. With a username the Pi logs in (587 STARTTLS or 465 SSL/TLS) — the mail then gets your domain's DKIM signature and passes DMARC. Without a username it is handed over without login, for a server that relays for the Pi by its address; its DKIM signer and SPF must know that address too. Test login only logs in, it sends nothing.
- E-mail: subject and text with placeholders {CALL} {NAME} {MY} {OP} {DATE} {UTC} {BAND} {MODE} {RST} {FREQ}. Title colour sets the mail's title bar (the gradient and edges follow from it; STEEL resets it). Preview mail opens the mail as the recipient sees it, with the form as it stands — sample QSO, nothing saved or sent. Send test card mails a card for a sample QSO to the test recipient.
- Sending: mark QSOs in Recent QSOs and click Send QSL. The address comes from QRZ.com (XML lookup, Settings ▸ Log). For one QSO the confirmation shows it, editable, with QRZ.com's QSL hints (eQSL / LoTW / paper and the operator's QSL note). For several, one confirmation; stations without an address, those that want no QSL and QSOs that already got a card are skipped and named.
- One card per station, band and mode and UTC day: AUTO QSL and several marked QSOs skip a station that already got its card today on that band and mode (QSL column ⊘); a single card sent by hand still goes, after a warning.
- No QSL wanted: a QSL note on QRZ.com such as NO QSL, NO EQSL, NO CARDS, LOTW ONLY, QRZ ONLY or OQRS ONLY, or eQSL: no. Such stations are skipped when several QSOs are marked and by AUTO QSL; for a single QSO the confirmation warns and leaves the choice to you.
- AUTO QSL (LOG title, or Settings ▸ QSL ▸ Automation) sends the card after every successful log, in the background on the Pi — no browser needed.
- What stays on the Pi only: the mail account with its password, the pictures and the title plate (backend/qsl_images/, runtime.json, backend/app/assets/ — all gitignored).

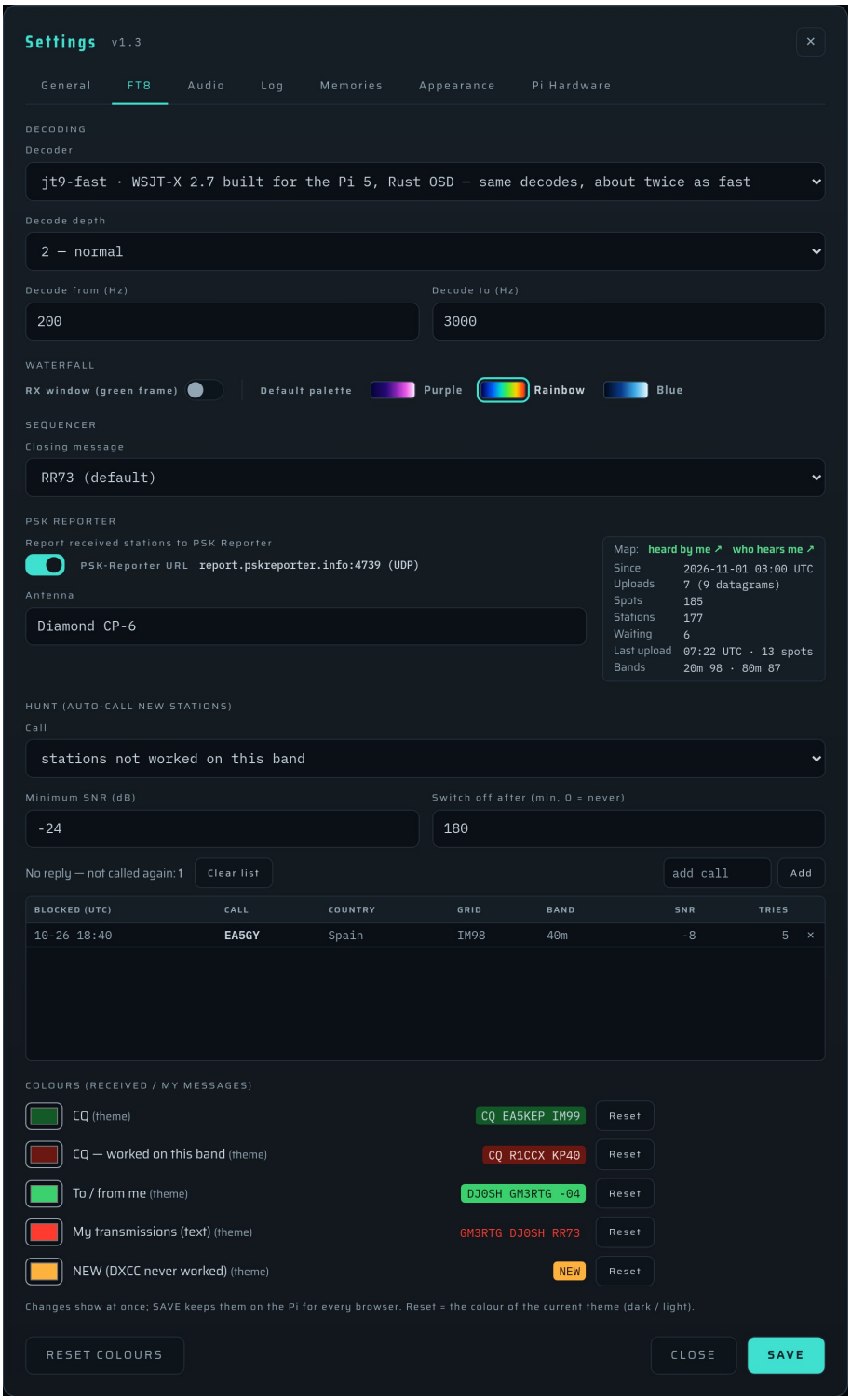
16 Settings

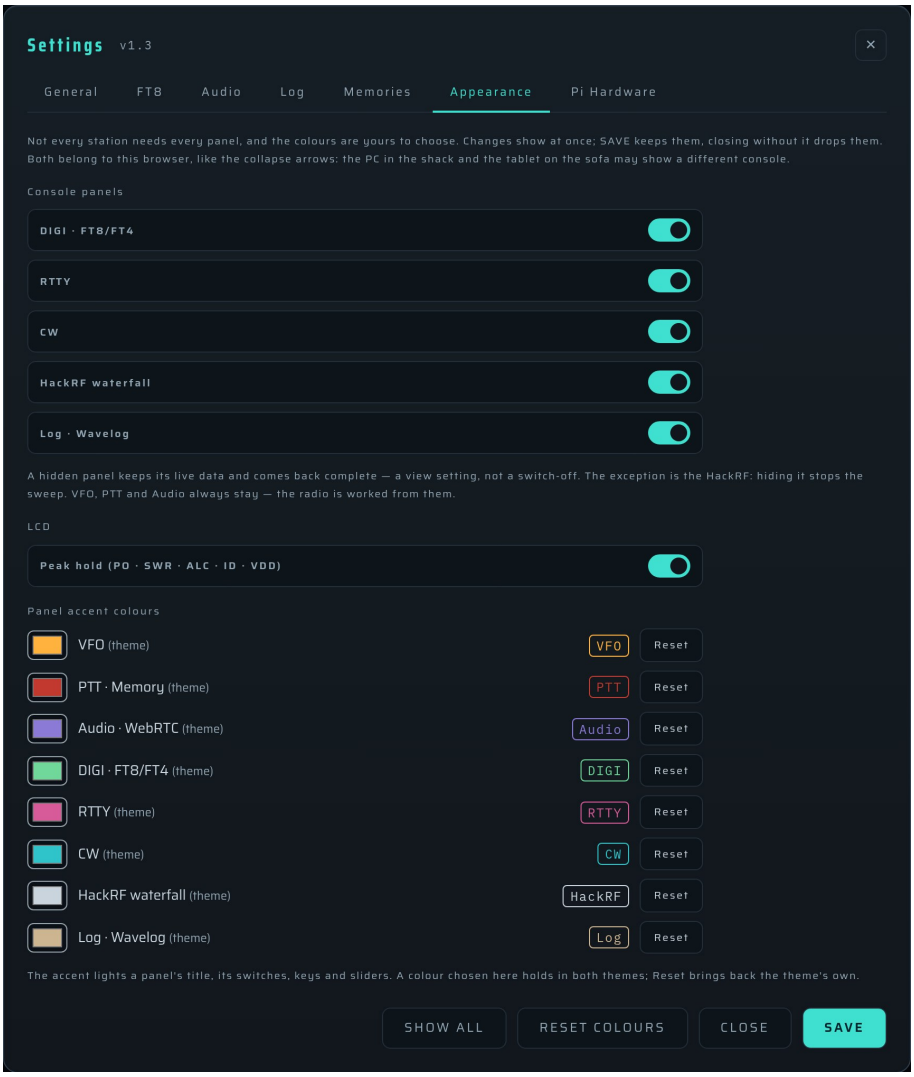


- General: callsign, locator, CAT port and baud rate, Shelly and auto power-off, HackRF source, logo, root CA.
- FT8: the decoder — jt9 (WSJT-X 2.7, Debian package), jt9-fast (the same decoder built for the Pi 5 with its OSD stage in Rust: identical decodes, about twice as fast) or jt9-33 (WSJT-X 3.3 beta, multithreaded MTft8 with a Rust OSD: about 6 % more decodes, ready about 2.4 s after the period; it sets its own depth). Builds not installed are greyed out. Then decode depth (as D1/D2/D3/AUTO in the panel) and range, closing message RR73 or RRR, PSK Reporter (switch, off by default, with the destination report.pskreporter.info:4739 (UDP) beside it; an optional antenna description; links to the PSK Reporter map — stations heard by you and stations that hear you, last hour; statistics since the service started: uploads, spots, different stations, waiting, last upload, spots per band. Uploads every 5 minutes, each station at most once an hour per band and mode, sooner only when its locator becomes known; unsure decodes and your own call are not reported), Auto Hunt (whom to call: not worked on this band, never worked, DXCC new on this band, DXCC never worked; minimum SNR; switch-off time, 0 = never; the no-reply list as a table — blocked time, call, country, grid, band, SNR, tries — with fixed column heads, × to take a call off, a field to add one by hand

and Clear list; these edits apply at once), the waterfall (per browser, at once: the green RX window on/off — default off —, the callsign strip for all palettes and the default palette, picked by clicking its colour sample; the Palette choice under the waterfall then lasts only until the page is reloaded), and the colours of the message lists with preview and reset.

- Audio: capture device, codec level, FT8 TX drive, the FT-950 audio menus, voice TX processing, WebRTC buffers.
- Log: Wavelog and QRZ.com; automation: QRZ lookup before every log, auto-log to Wavelog (as Auto-log in Sessions) and, separately, auto-log to the QRZ.com logbook (needs its API key).
- QSL: card picture and layout, mail server, e-mail text, test card, automatic sending (chapter 15).
- Memories: the 32 memory keys with CTCSS, split and VFO B.
- Appearance (per browser): hide the DIGI, RTTY, JTTY, CW, HackRF and Log panels (VFO, PTT and Audio always stay; hiding the HackRF stops its sweep), an accent colour per panel with Reset, peak hold of the LCD meters (PO · SWR · ALC · ID · VDD) on/off, and the AFU lamp on/off with its frequency plan. Changes show at once and are kept with SAVE. SAVE AS DEFAULT stores this browser's accent colours on the Pi as the default for every browser — a browser's own colours still win.
- Pi Hardware: load, temperature, memory, disk and a 24 h graph; below it the WSJT-X decoder (jt9) on its own — the last pass (CPU seconds, run time, parts), its memory (the parallel parts added up, peak) and the CPU of the last minute as a share of the Pi, with its own 24 h graph of CPU share and memory. CPU time comes from the kernel when a jt9 run ends; the memory is jt9's own peak, read while it runs. Two rows count audio dropouts since the service started: Audio · sound card shows the codec's overruns (samples lost — the decoder, waterfall and browser all miss them), slow 20 ms chunks (more than 15 ms to process) with the longest one, and arecord restarts; Audio · to browser shows gaps (a browser's buffer ran dry: a short silence), drops (a browser fell behind and the oldest audio was discarded) and how many browsers are listening. All at 0 means clean audio.


Settings v1.3
General
FT8
Audio
Log
Memories
Appearance
Pi Hardware
DECODING
Decoder
jt9-fast · WSJT-X 2.7 built for the Pi 5, Rust OSD — same decodes, about twice as fast
Decode depth
2 — normal
Decode from (Hz)
200
Decode to (Hz)
3000
WATERFALL
RX window (green frame)
Default palette
Purple
Rainbow
Blue
SEQUENCER
Closing message
RR73 (default)
PSK REPORTER
Report received stations to PSK Reporter
PSK-Reporter URL report.pskreporter.info:4739 (UDP)
Antenna
Diamond CP-6
Map: heard by me ↗ who hears me ↗
Since 2026-11-01 03:00 UTC
Uploads 7 (9 datagrams)
Spots 185
Stations 177
Waiting 6
Last upload 07:22 UTC · 13 spots
Bands 20m 98 · 80m 87
HUNT (AUTO-CALL NEW STATIONS)
Call
stations not worked on this band
Minimum SNR (dB)
-24
Switch off after (min, 0 = never)
180
No reply — not called again: 1
Clear list
add call
Add
BLOCKED (UTC)
CALL
COUNTRY
GRID
BAND
SNR
TRIES
10-26 18:40
EA5GY
Spain
IM98
40m
-8
5
COLOURS (RECEIVED / MY MESSAGES)
CQ (theme)
CQ EA5KEP IM99
Reset
CQ — worked on this band (theme)
CQ R1CCX KP40
Reset
To / from me (theme)
DJ0SH GM3RTG -04
Reset
My transmissions (text) (theme)
GM3RTG DJ0SH RR73
Reset
NEW (DXCC never worked) (theme)
NEW
Reset
Changes show at once; SAVE keeps them on the Pi for every browser. Reset = the colour of the current theme (dark / light).
RESET COLOURS
CLOSE
SAVE



Every tab ends with the same bar: CLOSE, plus SAVE on General, FT8, Log, QSL and Appearance. On Audio changes apply at once, on Memories each row has its own WRITE, Pi Hardware is read only. While Settings is open, the animations behind it pause (cycle bar, meters, spectrum, waterfalls) to save the browser's CPU; data and decoding go on, and the waterfalls draw what came in on close.

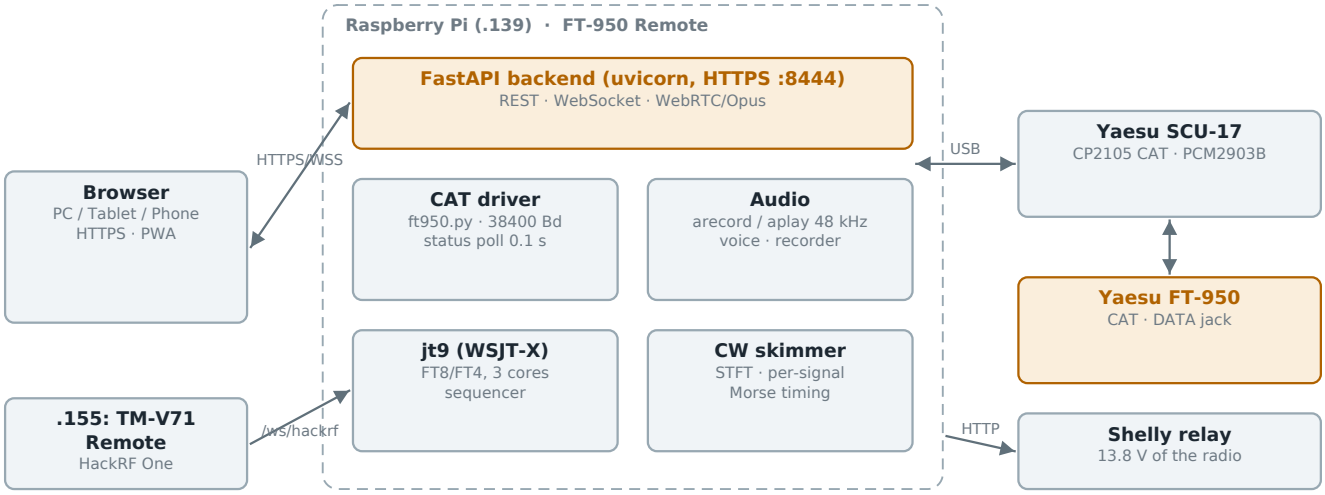
17 Remote power

A Shelly relay (Gen1 or Gen2/Plus/Pro, detected automatically) switches the radio's 13.8 V. The power switch in the title bar turns it on and off — before cutting the supply, a transmission and the sequencer are stopped. After switching off, power stays off for at least 3 s. While the radio is off a plate covers the page. Auto power-off switches the radio off when no browser has had the page open for the set time.

18 Troubleshooting

- CAT offline: check the by-id port and menu 026 = 3 (38400 Bd).
- No receive audio: CONNECT, then check RX GAIN and that MUTE/DROP are off.
- Nothing decoded: NTP must be synchronised (NTP lamp), mode DATA-U, the DIGI level in the green range.
- FT8 too quiet: FT8 TX drive (Settings ▸ Audio) and menu 050.
- No microphone: the page must be opened over HTTPS with the trusted CA.
- After a backend restart: reload the page and CONNECT again.
- No band strip under the waterfall: apt install voacapl, and set a locator in Settings ▸ General.

19 Technical reference



Backend

- FastAPI on uvicorn, systemd unit with RuntimeDirectory, StateDirectory, memory limits and Restart=always.
- ft950.py: CAT driver with read-back verification; radio_service.py polls frequency/mode/meters every 0.1 s and the receive chain every second.
- digi.py: UTC-anchored slices, jt9 in parallel sub-bands, adaptive depth, early and partner passes; sequencer.py: the QSO state machine.
- voice.py: WebRTC hub, PTT lease, pre-roll, AGC, compressor, pre-emphasis.
- cw.py: the CW skimmer; rtty.py: the RTTY skimmer and AFSK; jtty.py with decoder/jttyd: JTTY receive and transmit; dec_history.py: the decoder figures; hackrf_proxy.py; shelly.py; recorder.py.
- propagation.py: VOACAP band conditions — the ring of 36 paths, one run per UTC day, the day/night halves; GET /api/propagation.

Files on the Pi

```
backend/runtime.json      station data and credentials (0600, not in git)
backend/ex_defaults.json  saved FT-950 audio-menu defaults
backend/certs/            server certificate (from the private CA)
/run/ft950                decoder slices (tmpfs)
/var/lib/ft950            jt9 working data
backend/itshfbc/          VOACAP working tree (makeitshfbc copy)
backend/propagation.json  last sunspot number from NOAA
```

Source, issues and updates: github.com/CQ-DJ0SH/ft950-remote — GPL v3.